

Bayes factor

Readings for today

- Wagenmakers, E. J. (2007). A practical solution to the pervasive problems of p values. *Psychonomic bulletin & review*, 14(5), 779-804.

Topics

1. Problems with p-values

2. Bayes factors

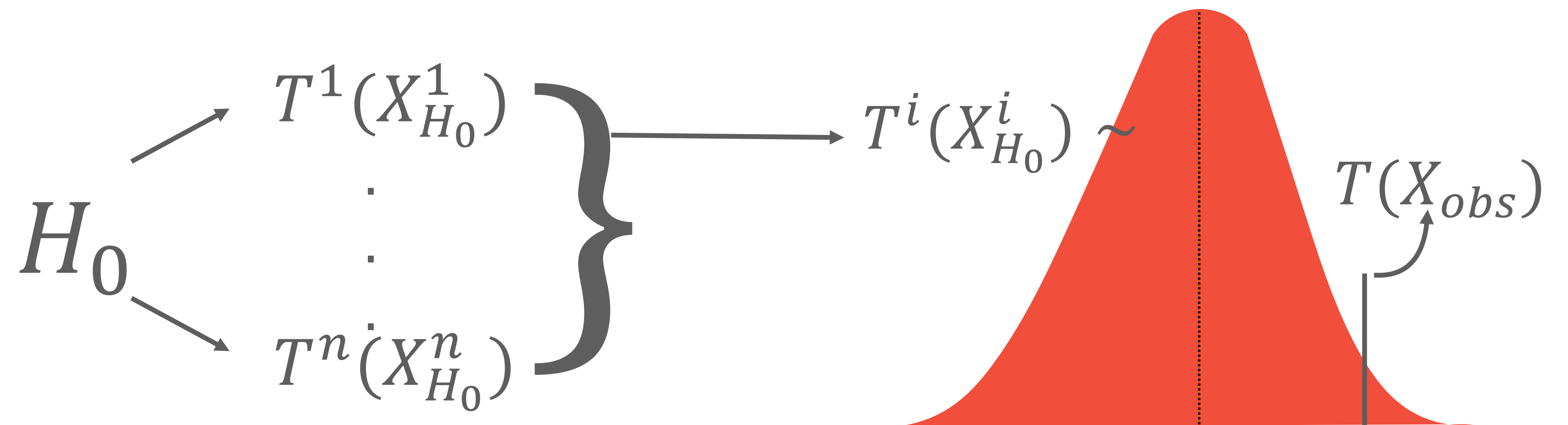
Problems with p-values

Null hypothesis test statistics (NHTSs)

$$p = P(Y|H_0 = \text{true}) = P(T(X_{H_0}) \geq T(X_{obs}))$$

observed data null hypothesis predictor variables generated by H_0 predictor variables you observe

test statistic

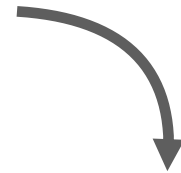


Problems with p-values

Fundamental limitations:

1. Depend on unobserved data.
2. Depend on subjective intentions.
3. Do not quantify statistical evidence.
4. Poorly understood.

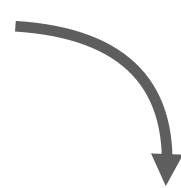
probability that the
null *exists*.


$$p = P(Y|H_o = \textit{true})$$

Ideal measure of hypotheses

Ideal traits of hypothesis evaluation:

1. Depends only on observed data.
2. Is independent of subjective intentions.
3. Is a measure of evidence, not existence.
4. Is easy to interpret.
5. Is objective.

Want 

evidence
 $\hat{=}$

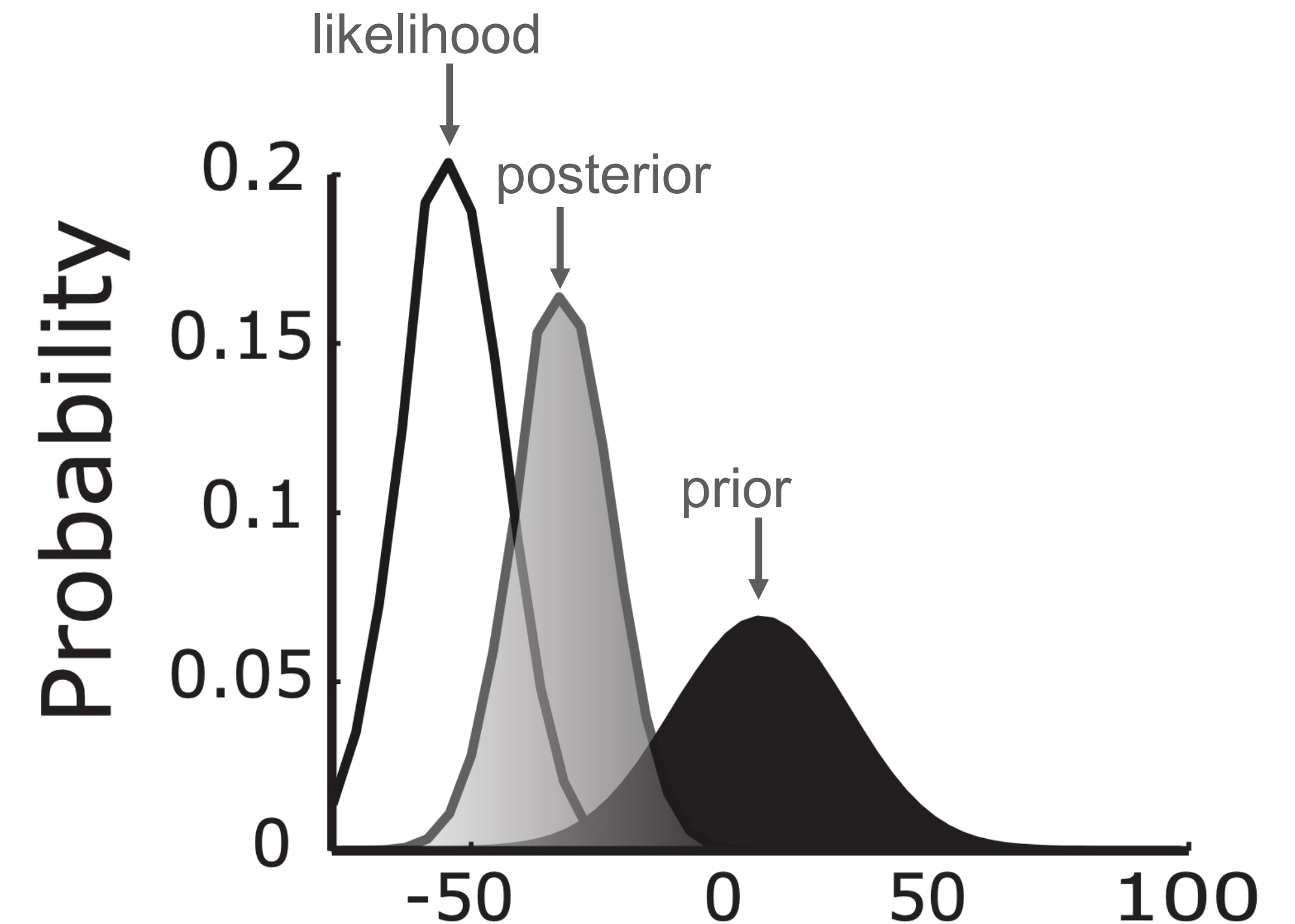
existence
 $\hat{=}$

$P(H_o = true|Y)$ not $P(Y|H_o = true)$

Bayes factor

Bayes Theorem

$$\underbrace{P(H_i|Y)}_{\text{posterior}} = \frac{\underbrace{P(Y|H_i)}_{\text{likelihood}} \underbrace{P(H_i)}_{\text{prior}}}{\underbrace{P(Y)}_{\text{marginal}}}$$



Example: (Gaussian) linear model

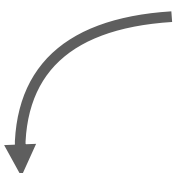
model: $\hat{f}(X) = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j X_j$

likelihood: $P(Y|\hat{f}(X)) = P(Y|X, \hat{\beta}) = \sum_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y_i - \hat{\beta}_0 - \sum_{j=1}^p \hat{\beta}_j x_{i,j})^2}{2\sigma^2}}$

prior: $P(\hat{f}(X)) = P(X, \hat{\beta}) = U(\underbrace{-\infty, +\infty}_{\text{unknown}}) \text{ or } N(\underbrace{\mu, \sigma}_{\text{known}})$

posterior: $P(\hat{f}(X)|Y) = -\frac{n}{2}\log(2\pi) - n\log(\sigma) - \frac{\sum_{i=1}^n (y_i - \hat{\beta}_0 - \sum_{j=1}^p \hat{\beta}_j x_{i,j})^2}{2\sigma}$

in log form



Bayes Factor

What is the ratio of evidence for two hypotheses?

$$BF_{01} = \frac{P(H_0|Y)}{P(H_1|Y)} = \frac{P(Y|H_0)}{P(Y|H_1)} \cdot \frac{P(H_0)}{P(H_1)}$$

 Bayes Factor for
 H_0 (against H_1)

- Determine the relative evidence for one hypothesis against the other.
- BF_{ij} identifies whether the observed data are more likely to arise from hypothesis i (H_i) than from hypothesis j (H_j).

Example: (Gaussian) linear model

$$\begin{aligned} \underline{H_1}: \hat{f}(X) &= \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 \\ \underline{H_0}: \hat{f}(X) &= \hat{\beta}_0 + \hat{\beta}_1 X_1 \end{aligned}$$

confound → X_1 (in H_1)
target → X_2 (in H_1)
confound → X_1 (in H_0)

Bayesian Information Criteria (BIC)

$$BIC(H_i) = p_i \log(n) - 2 \log(L_i)$$

parameters → p_i
 $L_i = \frac{RSS_i}{TSS}$

Difference in BICs

$$\begin{aligned} \Delta BIC_{10} &= BIC(H_1) - BIC(H_0) \\ &= n \log\left(\frac{RSS_1}{RSS_0}\right) + (p_1 - p_0) \log(n) \end{aligned}$$

$H_1 - H_0$ → ΔBIC_{10}

Bayes Factor for H_0

$$BF_{01} \propto \frac{P(Y|H_0)}{P(Y|H_1)} = e^{\frac{\Delta BIC_{10}}{2}}$$

Bayes Factor

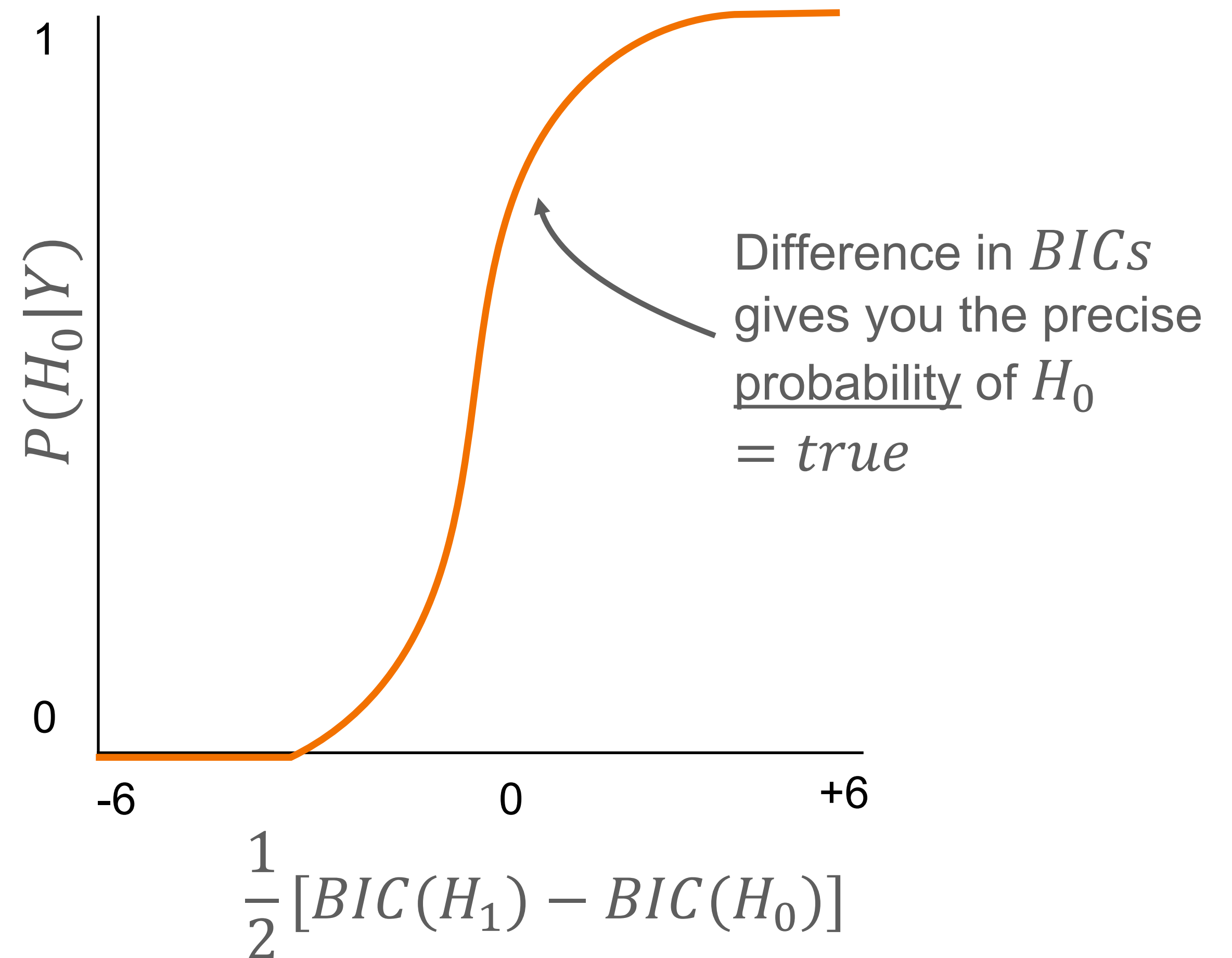
General formulation of BFs: Evaluate k-many alternative models

One against many

$$P(H_i|Y) = \frac{e^{-\frac{BIC(H_i)}{2}}}{\sum_{j=0}^{k-1} e^{-\frac{BIC(H_j)}{2}}}$$

H_0 alone

$$P(H_0|Y) = \frac{1}{1 + e^{-\frac{\Delta BIC_{10}}{2}}}$$



Inferring from BFs

No equivalent of $p < 0.05$ for BFs, so have to make inferential heuristics based on the strength of evidence.

BF_{01}	$P(H_0 Y)$	Evidence
1-3	0.50-0.75	weak
3-20	0.75-0.95	positive
20-150	0.95-0.99	strong
>150	>0.99	very strong

Take home message

- Bayes Factors offer the ability to directly estimate the evidence for one hypothesis over another. This is incredibly useful for making inferences for or against the null hypothesis itself.